

1 REFLX: REACTION ENHANCEMENT IN FITNESS
2 USING LIGHT-BASED EXERCISES FOR UPV ATHLETES

3 A Special Problem
4 Presented to
5 the Faculty of the Division of Physical Sciences and Mathematics
6 College of Arts and Sciences
7 University of the Philippines Visayas
8 Miag-ao, Iloilo

9 In Partial Fulfillment
10 of the Requirements for the Degree of
11 Bachelor of Science in Computer Science by

12 CELESTIAL, Christian Dale P.
13 YEBAN, Mark Rensel R.

14 Francis DIMZON, Ph.D.
15 Adviser

16 February 13, 2026

Approval Sheet

The Division of Physical Sciences and Mathematics, College of Arts and
Sciences, University of the Philippines Visayas

certifies that this is the approved version of the following special problem:

**REFLX: REACTION ENHANCEMENT IN FITNESS
USING LIGHT-BASED EXERCISES FOR UPV ATHLETES**

Approved by:**Name****Signature****Date**

Francis D. Dimzon, Ph.D.

(Adviser)

Ara Abigail E. Ambita

(Panel Member)

Christi Florence C. Cala-or

(Panel Member)

Kent Christian A. Castor

(Division Chair)

25 Division of Physical Sciences and Mathematics

26 College of Arts and Sciences

27 University of the Philippines Visayas

28 **Declaration**

29 We, Christian Dale P. Celestial and Mark Rensel D. Yeban, hereby certify that
30 this Special Problem has been written by us and is the record of work carried out
31 by us. Any significant borrowings have been properly acknowledged and referred.

Name

Signature

Date

32 Christian Dale P. Celestial

(Student)

Mark Rensel D. Yeban

(Student)

Dedication

“Hello, world.”

Acknowledgment

“Hello, world.”

Abstract

From 150 to 200 words of short, direct and complete sentences, the abstract should be informative enough to serve as a substitute for reading the entire SP document itself. It states the rationale and the objectives of the research. In the final Special Problem document (i.e., the document you'll submit for your final defense), the abstract should also contain a description of your research results, findings, and contribution(s).

Suggested keywords based on ACM Computing Classification system can be found at https://dl.acm.org/ccs/ccs_flat.cfm

Keywords: Keyword 1, keyword 2, keyword 3, keyword 4, etc.

47

Contents

48	1 Introduction	1
49	1.1 Overview of the Current State of Technology	1
50	1.2 Problem Statement	4
51	1.3 Research Objectives	4
52	1.3.1 General Objective	4
53	1.3.2 Specific Objectives	4
54	1.4 Scope and Limitations of the Research	5
55	1.5 Significance of the Research	5
56	2 Review of Related Literature	7
57	2.1 Theme 1 Title	8
58	2.2 Theme 2 Title	9

59	2.3 Chapter Summary	9
60	3 Research Methodology	11
61	3.1 Research Activities	11
62	4 Results and Discussions	13
63	5 Conclusion	15
64	6 References	17
65	A Code Snippets	19
66	B Resource Persons	21

67 List of Figures

68	1.1	This is the figure’s caption – Disney stock chart. Captions should	
69		fully describe the figure in a concise manner such that there is no	
70		need to refer to the text when figuring out the graphic.	2

⁷¹ List of Tables

Chapter 1

Introduction

1.1 Overview of the Current State of Technology

This section gives the reader an overview of the specific technology or field in the international or local setting. The information regarding the technology or field should be contemporary and not based on outdated sources. Discussion must not be too technical or too detailed.

This section ends with a discussion on the problem/s faced by or that still exist in the specific technology or field (e.g., limitations of existing software or algorithms). The problem statement would lead to the research objectives.

It is easy to include a figure in JPG or PNG format as shown in the following example. Make sure that you explain what the figure is all about, and that you refer to your figure. Figures and Tables should appear after they were referred to in the text. For example, Figure 1.1 shows a graph of the performance of Disney



Figure 1.1: This is the figure’s caption – Disney stock chart. Captions should fully describe the figure in a concise manner such that there is no need to refer to the text when figuring out the graphic.

86 stock from the 1980s to 2012.

87 Some notes on citing references. When using APA format, the author-date method
 88 of citation is followed. This means that the author’s last name and the year of
 89 publication for the source should appear in the text, and a complete reference
 90 should appear in the reference list.

91 Here are some examples on how to do the referencing (note author’s name and
 92 years are different from commented examples). For APA citation details, refer to
 93 <http://www.ctan.org/tex-archive/biblio/bibtex/contrib/apacite/>.

94 • ? (?) compared reaction times...

95 • In a recent study of reaction times (?, ?)...

96 • In ?, ? compared reaction times...

97 • ? (?) compared reaction times...

98 • In a recent study of reaction times (?, ?)...

99 • In ?, ?, compared reaction times...

100 The following are references from journal articles (?, ?, ?, ?). Here's an MS thesis
101 document (?, ?), and this is from PhD dissertation (?, ?). For a book, reference
102 is given as (?, ?). Proceedings from a conference samples are (?, ?, ?, ?). The
103 sample bibliography file named **myreferences.bib** is from the SIGGRAPH L^AT_EX
104 template. You can use a text editor to view the contents of the bib file. It is your
105 task to create your own bibliography file. For those who downloaded papers from
106 ACM or IEEE sites, there is a BibTeX link that you can click; thereafter, you just
107 simply need to copy and paste the BibTeX entry into your own bibliography file.

108 The following shows how to include a program source code (or algorithm). The
109 verbatim environment, as the name suggests, outputs text (including white spaces)
110 as is...

```
111           #include <stdio.h>
112           main()
113           {
114                 printf("Hello world!\n");
115           }
```

1.2 Problem Statement

DO NOT FORGET to write the statement of the research problem here, i.e., before the Research Objectives.

A problem statement is your research problem written explicitly. The problem statement should do four things:

1. Specify and describe the problem (with appropriate citations)
2. Provide evidence of the problem's existence
3. Explain the consequences of NOT solving the problem
4. Identify what is not known about the problem that should be known.

1.3 Research Objectives

1.3.1 General Objective

This subsection states the over-all goal that must be achieved to answer the problem. Address the following: Given your research challenge or opportunity, how do you intend to solve it? What is the output of your research?

1.3.2 Specific Objectives

This subsection is an elaboration of the general objective. It states the specific steps that must be undertaken to accomplish the general objective. These objec-

133 tives must be **S**pecific, **M**easurable, **A**ttainable, **R**ealistic, **T**ime-bounded. A spe-
134 cific objective start with “to <verb>” for example: to design/survey/review/analyze.

135 Studying a particular programming language or development tool (e.g., to study
136 Windows/Object-Oriented/Graphics/C++ programming) to accomplish the gen-
137 eral objective is inherent in all thesis and, therefore, must not be included here.

- 138 1. To review related literature, compare and contrast existing algorithms (on
139 what problem?);
- 140 2. To develop a new algorithm (for what purpose?)
- 141 3. To analyze the algorithm (based on what criteria?)

142 1.4 Scope and Limitations of the Research

143 This section discusses the boundaries (with respect to the objectives) of the re-
144 search and the constraints within which the research was developed.

145 1.5 Significance of the Research

146 This section explains why research was done in this area. It rationalizes the ob-
147 jective of the research with that of the stated problem. Avoid including sentences
148 such as “This research is beneficial to the proponent/department/college” as this
149 is already an inherent requirement of all BSCS majors. Focus on the research’s
150 contribution to the Computer Science field.

151 The following are guide questions that may help your formulate the significance
152 of your research.

- 153 • What is the relevance of your work to the computer science community?
 - 154 – What are your technical contributions, in terms of algorithms, or ap-
155 proaches, or new domain?
 - 156 – What is your value-added compared to existing systems?
- 157 • What are your contributions to society in general?
 - 158 – Who benefits from your system?
 - 159 – Who are your target users and how this system benefit them?

Chapter 2

Review of Related Literature

This chapter discusses the features, capabilities, and limitations of existing research, algorithms, or software that are related/similar to the Special Problem.

The reviewed works and software must be arranged either in chronological order, or by area (from general to specific). Observe a consistent format when presenting each of the reviewed works. This must be selected in consultation with the adviser.

DO NOT FORGET to cite your references.

A literature review must do these things:

- be organized around and related directly to the thesis or research question you are developing
- synthesize results into a summary of what is and is not known
- identify areas of controversy in the literature

- 173 • formulate questions that need further research

174 A literature review is a piece of discursive prose, not a list describing or summa-
175 rizing one piece of literature after another. It's usually a bad sign to see every
176 paragraph beginning with the name of a researcher. Instead, organize the litera-
177 ture review into sections that present themes or identify trends, including relevant
178 theory. You are not trying to list all the materials published, but to synthesize
179 and evaluate them according to the guiding concept of your thesis or research
180 question. You should also state the limits or gaps of their researches wherein you
181 will try to fill these gaps in accordance to your research problem and objectives.

182 2.1 Theme 1 Title

183 This chapter contains a review of research papers that:

- 184 • Describes work on a research area that is similar or relevant to yours
- 185 • Describes work on a domain that is similar or relevant to yours
- 186 • Uses an algorithm that may be useful to your work
- 187 • Uses a software / tool that may be useful to your work

188 It also contains a review of software systems that:

- 189 • Belongs to a research area similar to yours
- 190 • Addresses a need or domain similar to yours
- 191 • Is your predecessor

192 **2.2 Theme 2 Title**

193 **2.3 Chapter Summary**

194 Should include a table of related studies comparing them based on several criteria.

195 Highlight research gaps and the research problem.

Chapter 3

Research Methodology

This chapter lists and discusses the specific steps and activities that were performed to accomplish the project. The discussion covers the activities from pre-proposal to Final SP Writing.

3.1 Research Activities

Research activities include inquiry, survey, research, brainstorming, canvassing, consultation, review, interview, observe, experiment, design, test, document, etc. Be sure that for each method, process, or algorithm used, there is a justification why that method was chosen. The methodology also includes the following information:

- who is responsible for the task
- the resource person to be contacted

- 209 • what were done
- 210 • when and how long the activity was done
- 211 • where it was done
- 212 • why should the activity was done

213 **Chapter 4**

214 **Results and Discussions**

215 This chapter presents the results or the system of your SP. Include screenshots,
216 tables, or graphs and provide the discussion of results.

217 **Chapter 5**

218 **Conclusion**

219 This chapter summarizes your SP and provides conclusions regarding your results
220 and analyses. Provide recommendations on what ought to be done with your SP
221 or provide further directions on the topic you covered.

²²² Chapter 6

²²³ References

²²⁴ **Appendix A**

²²⁵ **Code Snippets**

226 **Appendix B**

227 **Resource Persons**

228 **Dr. Firstname1 Lastname1**

229 Role1

230 Affiliation1

231 emailaddr@domain.com

232 **Mr. Firstname2 Lastname2**

233 Role2

234 Affiliation2

235 emailaddr2@domain.com

236 **Ms. Firstname3 Lastname3**

237 Role3

238 Affiliation3

239 emailaddr3@domain.net

240